INITIAL RESEARCH PROPOSAL

Relevance-Aware Interruption and Re-entry Support for Partial Participation in Remote Meetings

Group number: _____ Members / student IDs: _____

1. Background

Remote meetings may overlap with classes, other meetings, or focused work. Partial participation allows people to follow relevant developments and contribute without remaining continuously engaged. Bai et al. [1] implemented ProxyMe, which combines large language model (LLM)-generated updates with questions deferred to the user. Their 24-participant study compared update and response styles and identified trade-offs between awareness, interruption, and control, rather than one universally optimal setup.

Our classroom formative survey (seven respondents) provides an initial motivation [2]. All seven expressed concern about missing personally relevant information and moments requiring their decisions. Six selected direct mentions and six selected requests for their decision/opinion as deserving immediate alerts; only one selected general progress updates. All seven wanted configurable priorities/keywords, and six wanted a quick recap of the previous 1–3 minutes. However, four reported no such meeting/task conflict in the preceding six months. These small-sample, self-reported preferences guide design; they do not establish general demand or system effectiveness.

2. Problem

Fixed update schedules alone do not specify which events justify interrupting a particular user. An alert without context leaves users to reconstruct the discussion. Building on ProxyMe's suggestions on urgency and user control [1], we study relevance-aware routing, with context packs supporting re-entry. Our initial target is university project-team members with recent overlapping commitments.

RQ1. Compared with periodic updates and immediate alerts for all user-directed questions, how does relevance-aware routing affect interruption burden and timely, valid responses to decision-critical events?

RQ2. What context and interaction choices do participants use when deciding whether to reply, seek clarification, briefly join, or defer, and why?

3. Expected Method

3.1 Refine requirements

We plan 5-6 semi-structured interviews with people who have experienced meeting/task conflicts in the past six months. Recent examples and scenario cards will clarify immediate-alert criteria and re-entry needs. We will group responses into themes, then freeze the priority rules and context fields before evaluation.

3.2 Build a scoped prototype

We tentatively plan a desktop Unity simulator with a simple 3D meeting room and a Python/LLM backend. Fictional, time-stamped transcripts, speaker labels, agendas, and role profiles replace live conferencing. Brief joining triggers scripted dialogue and acknowledgements; later meeting events remain fixed. VR hardware and production integration are outside the initial scope.

Using only transcript content available so far, the backend labels mentions, input requests, responsibility changes, and routine discussion. A decision-critical event requires the assigned user's input before a scripted deadline, even without an explicit user-directed question; a mention alone is insufficient. The prototype supports both fixed routing rules defined in Section 3.3. Ambiguous outputs keep source excerpts, and classification failures are logged. Context packs show what happened, why input is needed, the pending question, and supporting excerpts. Users may edit/type replies, request context, briefly join, defer, or pause. No reply is relayed without explicit confirmation.

3.3 Evaluate the policy and interaction

First, two team members will independently label held-out scripts for role relevance, decision dependencies, and deadlines. Event rules and a role-based response rubric will be fixed, with disagreements resolved, before viewing model output. The LLM test will report false alerts, missed critical events, classification failures, and event-to-alert latency. Next, a counterbalanced within-participant pilot targets 12-16 eligible participants. Each completes reading tasks during two matched 10-minute meetings. Both meetings include explicit user-directed questions and implicit decision-critical events. Order and scenario assignments are balanced; critical-event counts, timing, and reading difficulty are matched.

The user study applies both policies to frozen, human-labelled event streams, separating routing effects from LLM errors. Our ProxyMe-inspired baseline (A) alerts for every user-directed question; relevance-aware routing (B) alerts for critical events and fixed role priorities. Every event is delivered once, immediately or in the shared 60-second digest. Both use identical content fields, detail levels, role inputs, and reply controls, with records available on demand; only routing differs. This is not a ProxyMe replication or a live-system effectiveness test.

The focal outcomes are immediate-alert count and the proportion of all critical events receiving a rubric-valid confirmed reply before their deadline. Other outcomes are timely invalid reply, late reply, technical failure, deliberate deferral, pause, or no response; none leave the denominator. Confirmed replies take precedence; otherwise we record failure, the final defer/pause choice, or no response. Event-to-confirmation latency is descriptive for confirmed replies only. Comprehension quizzes, the Rating Scale Mental Effort (RSME), and a 7-point perceived-interruption rating provide supporting measures. We will describe paired differences with uncertainty estimates; this exploratory pilot cannot establish non-inferiority. Observations and interviews address RQ2. Fewer alerts alone are insufficient if critical participation deteriorates; context-card effects are not isolated.

3.4 Consent, scope, and expected output

Studies will use fictional meetings, informed consent, and anonymised logs; required course/university approval will precede recruitment. Expected outputs are a prototype and preliminary design guidance. Sample targets, Unity/VR requirements, and scope remain subject to both TAs' confirmation; we will report the achieved sample.

References

[1] Zhongyi Bai, Nadya Ee Png, and Eduardo Velloso. 2026. Enabling Partial Participation in Remote Meetings. Proceedings of CHI 2026, 20 pages. doi:10.1145/3772318.3790744.

[2] Project team. 2026. Partial Participation in Remote Meetings: Formative Survey Results on Awareness, Interruption, and Re-engagement. Unpublished AIAA 4435 classroom presentation (v3), slides 3–6.